PDS Search LDD Form

Scientific Discipline

Information about the scientific discipline and research context.

Primary Field Primary scientific field of study.

- ☐ **Astrobiology** Study of life in the universe
- ☐ **Astronomy Observation** Observation of celestial objects
- ☐ **Astronomy Radar** Radar observations of celestial objects
- ☐ **Astronomy Exoplanets** Study of exoplanets
- ☐ **Astronomy Spectroscopy** Spectroscopy of celestial objects
- ☐ **Geology Geophysics** Geophysics of the Earth and other planets
- ☐ **Geology Seismology** Seismology of the Earth and other planets
- ☐ **Geology Surface** Surface studies of the Earth and other planets
- ☐ **Geology Crater Counting** Crater counting of the Earth and other planets
- ☐ **Geology Interior Studies** Interior studies of the Earth and other planets
- ☐ **Dynamics** Dynamics of the Earth and other planets
- ☐ **Laboratory Geochemistry** Geochemistry of physical samples
- ☐ **Laboratory Cosmochemistry** Cosmochemistry of physical samples
- ☐ **Laboratory Age Dating** Age dating of physical samples
- ☐ **Laboratory Analytic Chemistry** Analytic chemistry of physical samples
- ☐ **Laboratory Spectral Analysis** Spectral analysis of physical samples
- ☐ **Atmospheric Study** Atmospheric studies of the Earth and other planets
- ☐ **Modeling** Modeling of the Earth and other planets
- ☐ **Magnetospheres** Magnetospheres of the Earth and other planets
- ☐ **Space Physics** Space physics of the Earth and other planets
- ☐ **Other** Any field not covered by other values

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Investigation Technique Method of scientific investigation used.

- ☐ **Spectroscopy** Collection of spectra
- ☐ **Modeling** Computer-based simulation or modeling
- ☐ **Mapping** Collection of maps
- ☐ **Radar** Collection of radar data
- ☐ **Seismology** Collection of seismic data
- ☐ **Gravimetry** Collection of gravity data
- ☐ **Geochemistry** Collection of geochemical data
- ☐ **SEM** Collection of scanning electron microscopy data
- ☐ **TEM** Collection of transmission electron microscopy data
- ☐ **ICPMS** Collection of inductively coupled plasma mass spectrometry data
- ☐ **Other** Any investigation technique not covered by other values

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Study Focus Specific area of study within the scientific discipline.

- ☐ **Surface** Study focused on surface features or properties
- ☐ **Interior** Study focused on internal structure or processes
- ☐ **Atmosphere** Study focused on atmospheric properties or dynamics
- ☐ **Magnetosphere** Study focused on magnetic field properties or interactions
- ☐ **Orbital Dynamics** Study focused on orbital properties or behaviors
- ☐ **Composition** Study focused on chemical or mineral composition
- ☐ **Other** Any focus not covered by other values

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Target Object

Information about the target of observation, whether it is a celestial body, feature, or other object.

Object Name Name of the object

Value: ______________________________

Parent Body Parent body of the object

Value: ______________________________

Scope Scope of the data collection

- ☐ **Global** Data collected from the entire object
- ☐ **Regional** Data collected from a large region of the object
- ☐ **Local** Data collected from a small location on the object
- ☐ **Other** Any scope not covered by other values

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Object Type

Type of the object

Planet Planet

- ☐ **Gas Giant** Large planets composed mainly of gas
- ☐ **Terrestrial** Rocky planets similar to Earth
- ☐ **Interior** Planets closer to the Sun than Earth
- ☐ **Other** Any planet type not covered by other values

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Fields Fields of an object

- ☐ **Particles** Particles in the atmosphere
- ☐ **Cosmic Rays** Cosmic rays in the atmosphere
- ☐ **X-Rays** X-rays in the atmosphere
- ☐ **Gamma-Rays** Gamma-rays in the atmosphere
- ☐ **Electromagnetic Waves** Electromagnetic waves in the atmosphere

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Dust Dust

- [ ] **Meteoroids** Dust from meteoroids
- [ ] **Other** Other dust

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Solar Component Solar component

- [ ] **Corona** Corona
- [ ] **Solar Wind** Solar Wind
- [ ] **Exoplanets** Exoplanets
- [ ] **Heliosphere** Heliosphere
- [ ] **Core** Core
- [ ] **Radiative Zone** Radiative Zone
- [ ] **Convective Zone** Convective Zone
- [ ] **Photosphere** Photosphere
- [ ] **Chromosphere** Chromosphere
- [ ] **Other** Any solar component not covered by other values

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Exoplanet Name of an exoplanet

Value: ______________________________

Small Body

Small body

Asteroid Asteroid

- [ ] **Near Earth** Asteroids with orbits that bring them close to Earth
- [ ] **Trojan** Asteroids that share an orbit with a planet
- [ ] **Active** Asteroids that show comet-like activity
- [ ] **Main Belt** Asteroids in the main asteroid belt
- [ ] **Other** Any asteroid type not covered by other values

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TNO Trans-Neptunian Object

- [ ] **Classical** Objects in the classical Kuiper belt
- [ ] **Resonant** Objects in orbital resonance with Neptune
- [ ] **Scattered** Objects scattered by Neptune
- [ ] **Detached** Objects with orbits not strongly influenced by Neptune
- [ ] **Other** Any TNO type not covered by other values

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Centaur Centaur

- [ ] **Active** Centaurs showing comet-like activity
- [ ] **Inactive** Centaurs without comet-like activity
- [ ] **Other** Any centaur type not covered by other values

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KBO Kuiper Belt Object

- ☐ **Classical** Objects in the classical Kuiper belt
- ☐ **Resonant** Objects in orbital resonance with Neptune
- ☐ **Scattered** Objects scattered by Neptune
- ☐ **Other** Any KBO type not covered by other values

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Oort Cloud Oort Cloud Object

- ☐ **Inner** Objects in the inner Oort cloud
- ☐ **Outer** Objects in the outer Oort cloud
- ☐ **Other** Any Oort cloud type not covered by other values

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Comet Comet

- ☐ **Jupiter Family** Comets that orbit Jupiter
- ☐ **Single Pass** Comets that pass through the inner solar system
- ☐ **Other** Any comet type not covered by other values

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Satellite Satellite

- ☐ **Icy Moons** Icy moons of a planet
- ☐ **Galilean Moons** Galilean moons of Jupiter
- ☐ **Irregular Moons** Irregular moons of a planet
- ☐ **Captured Moons** Captured moons of a planet
- ☐ **Shepherding Moon** Shepherding moon of a planet
- ☐ **Other** Any satellite type not covered by other values

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Feature Feature

Name Name of the feature

Value: ____

Type Type of the feature

- ☐ **Crater** Impact depression on a surface
- ☐ **Mountain** Elevated landform
- ☐ **Valley** Elongated depression
- ☐ **Mare** Dark plains on the Moon
- ☐ **Volcano** Volcanic landform
- ☐ **Cloud** Atmospheric cloud formation
- ☐ **Other** Any feature type not covered by other values

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Atmosphere

Atmosphere of a celestial body

Layer

Layer of the atmosphere

- ☐ **Troposphere** Lowest layer of the atmosphere
- ☐ **Stratosphere** Layer above the troposphere
- ☐ **Ionosphere** Layer containing ions
- ☐ **Exosphere** Outermost layer of the atmosphere
- ☐ **Other** Any atmospheric layer not covered by other values

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Study Type

Type of study of the atmosphere

- ☐ **Temperature Profile** Temperature profile of the atmosphere
- ☐ **Scale Height** Scale height of the atmosphere
- ☐ **Density Profile** Density profile of the atmosphere

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Rings

Rings of an object

Ring Type

Type of ring

- ☐ **D** D ring
- ☐ **C** C ring
- ☐ **B** B ring
- ☐ **A** A ring
- ☐ **F** F ring
- ☐ **G** G ring
- ☐ **E** E ring

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Ring Division

Division of the ring

- ☐ **Cassini Division** Cassini Division
- ☐ **Encke** Encke Division
- ☐ **Keeler** Keeler Division

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Ring Feature

Feature of the ring

- ☐ **Spiral Bending Waves** Spiral Bending Waves
- ☐ **Spokes** Spokes
- ☐ **Spiral Density Waves** Spiral Density Waves

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Physical Properties

Information about the physical properties of the object under study.

Material Type Material type of the object

- ☐ **Rock** Solid mineral material
- ☐ **Ice** Frozen water or other volatiles
- ☐ **Gas** Gaseous material
- ☐ **Dust** Fine particulate material
- ☐ **Regolith** Surface layer of loose material
- ☐ **Liquid** Liquid state material
- ☐ **Other** Any material type not covered by other values

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Process Type Type of process being studied.

- ☐ **Impact** Collision processes
- ☐ **Volcanic** Volcanic activity
- ☐ **Tectonic** Crustal deformation processes
- ☐ **Atmospheric** Atmospheric processes
- ☐ **Erosional** Surface degradation processes
- ☐ **Other** Any process type not covered by other values

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Composition Primary composition of the material being studied.

- ☐ **Silicate** Composed primarily of silicate minerals
- ☐ **Metal** Composed primarily of metallic elements
- ☐ **Organic** Composed primarily of organic compounds
- ☐ **Icy** Composed primarily of water or other volatile ices
- ☐ **Gaseous** Composed primarily of gas
- ☐ **Mixed** Composed of a mixture of different materials
- ☐ **Other** Any composition not covered by other values

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Environmental Context

Information about the environmental conditions at the study location.

Temperature Regime Temperature conditions at the study location.

- ☐ **Cryogenic** Extremely low temperatures, typically below 120K
- ☐ **Cold** Below freezing but not extreme
- ☐ **Temperate** Near-surface temperatures typical of Earth
- ☐ **Warm** Above freezing and below boiling point
- ☐ **Hot** Above boiling point
- ☐ **Other** Any temperature regime not covered by other values

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Pressure Conditions Pressure conditions at the study location.

- ☐ **Vacuum** Essentially no atmospheric pressure
- ☐ **Very Low** Minimal atmospheric pressure like Mars
- ☐ **Earth Like** Similar to Earth's atmospheric pressure
- ☐ **High** Higher than Earth's atmospheric pressure
- ☐ **Extreme** Very high pressure like Venus surface or gas giant interiors
- ☐ **Variable** Significant pressure variations or gradients
- ☐ **Other** Any pressure condition not covered by other values

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Radiation Environment Radiation conditions at the study location.

- ☐ **Low** Minimal radiation exposure like Earth surface
- ☐ **Moderate** Similar to Earth orbit or Mars surface
- ☐ **High** Significant radiation like Jupiter radiation belts
- ☐ **Extreme** Very high radiation levels
- ☐ **Shielded** Protected or shielded from external radiation
- ☐ **Other** Any radiation environment not covered by other values

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Atmospheric Presence Presence and nature of atmosphere at the study location.

- ☐ **None** No atmosphere present
- ☐ **Trace** Very low atmospheric pressure
- ☐ **Low** Atmospheric pressure similar to Earth's
- ☐ **High** Atmospheric pressure similar to Earth's
- ☐ **Extreme** Very high atmospheric pressure like Venus surface
- ☐ **Variable** Significant atmospheric pressure variations or gradients
- ☐ **Other** Any atmospheric presence not covered by other values

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Data Collection

Information about how the data was collected and its format.

Collection Method Method used to collect the data.

- ☐ **Orbiter** Data collected by an orbiting spacecraft
- ☐ **Lander** Data collected by a surface lander
- ☐ **Rover** Data collected by a mobile surface rover
- ☐ **Telescope** Data collected by Earth-based or space telescopes
- ☐ **Sample Return** Analysis of returned physical samples
- ☐ **Laboratory** Data collected in laboratory settings
- ☐ **Other** Any collection method not covered by other values

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Instrument Type   Type of instrument used to collect the data.

- [ ] **Spectrometer** Instrument that measures spectra
- [ ] **Camera** Imaging device
- [ ] **Radar** Radio detection and ranging instrument
- [ ] **Lidar** Light detection and ranging instrument
- [ ] **Seismometer** Instrument that measures ground motion
- [ ] **Magnetometer** Instrument that measures magnetic fields
- [ ] **Other** Any instrument type not covered by other values

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Data Category   General category of the collected data.

- [ ] **Altimetry** Elevation data
- [ ] **Image** Visual representation data
- [ ] **Photometry** Photometric measurement data
- [ ] **Map** Spatial representation data
- [ ] **Model Output** Results from computational models
- [ ] **Database** Flat rendering of existing data base
- [ ] **Parameter Data** Numerical data
- [ ] **Other** Any data category not covered by other values

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Data Format   Format of the collected data.

- [ ] **Shape Model DTM** A 3D model of the surface of the object
- [ ] **Geologic Map** A map of the geologic features of the object
- [ ] **Thermal Map** A map of the thermal properties of the object
- [ ] **Mosaic** A composite image of the object
- [ ] **Spectral Map** A map of the spectral properties of the object
- [ ] **Gazetteer** A list of feature names
- [ ] **Time Series** Sequential data collected over time
- [ ] **Crater** A list of craters on the object
- [ ] **Light Curve** A list of light curves on the object
- [ ] **Spectral** A list of spectra on the object
- [ ] **Physical Properties** A list of physical properties of the object
- [ ] **Photometric Properties** A list of photometric properties of the object
- [ ] **Gravity Map** A map of the gravity of the object
- [ ] **Taxonomies** A list of taxonomies of the object
- [ ] **Crater Type** A list of irregular crater types on the object
- [ ] **Other** Any data format not covered by other values

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Temporal Epoch   The temporal epoch of the data collection.

Earth Temporal epoch of the Earth

- ☐ **Archean** Earth's earliest known epoch
- ☐ **Proterozoic** Earth's Proterozoic Eon
- ☐ **Paleozoic** Earth's Paleozoic Era

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Moon Temporal epoch of the Moon

- ☐ **Nectarian** Moon's Pre-Nactarian Epoch

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Mars Temporal epoch of Mars

- ☐ **Noachian** Mars's Noachian Epoch
- ☐ **Hesperian** Mars's Hesperian Epoch
- ☐ **Amazonian** Mars's Amazonian Epoch

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Analysis Properties

Information about the analysis methods and properties of the data.

Analysis Method Method used to analyze the data.

- ☐ **Spectroscopy** Collection of spectra
- ☐ **Modeling** Computer-based simulation or modeling
- ☐ **Mapping** Collection of maps
- ☐ **Radar** Collection of radar data
- ☐ **Seismology** Collection of seismic data
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- ☐ **SEM** Collection of scanning electron microscopy data
- ☐ **TEM** Collection of transmission electron microscopy data
- ☐ **ICPMS** Collection of inductively coupled plasma mass spectrometry data
- ☐ **Other** Any analysis method not covered by other values

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Software Used Software used to analyze the data.

- ☐ **Spectroscopy** Collection of spectra
- ☐ **Modeling** Computer-based simulation or modeling
- ☐ **Mapping** Collection of maps
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- ☐ **Seismology** Collection of seismic data
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- ☐ **Geochemistry** Collection of geochemical data
- ☐ **SEM** Collection of scanning electron microscopy data
- ☐ **TEM** Collection of transmission electron microscopy data
- ☐ **ICPMS** Collection of inductively coupled plasma mass spectrometry data
- ☐ **Other** Any software not covered by other values

Processing Steps Processing steps used to analyze the data.

- [ ] **Filtering** Filtering of the data
- [ ] **Smoothing** Smoothing of the data
- [ ] **Deconvolution** Deconvolution of the data
- [ ] **Other** Any processing step not covered by other values

Uncertainty Measurements Uncertainty measurements of the data.

- [ ] **Standard Deviation** Standard deviation of the data
- [ ] **Standard Error** Standard error of the data
- [ ] **Confidence Interval** Confidence interval of the data
- [ ] **Other** Any uncertainty measurement not covered by other values

Quality Indicators Quality indicators of the data.

- [ ] **Accuracy** Accuracy of the data
- [ ] **Precision** Precision of the data
- [ ] **Other** Any quality indicator not covered by other values

Research Context

Information about the research context and associated metadata.

Research Goals Research goals of the study

- [ ] **Accuracy** Accuracy of the data
- [ ] **Precision** Precision of the data
- [ ] **Other** Any research goal not covered by other values

Related Publications Related publications of the study

Value: ______________________

Keywords Keywords of the study

Value: ______________________

Principal Investigator Principal investigator of the study

Value: ______________________

Institution Institution of the study

Value: ______________________